

**Analog Communication Laboratory**

**EXPERIMENT 4 : FREQUENCY MODULATION**

**INDIAN INSTITUTE OF TECHNOLOGY  
KHARAGPUR**



**DEPARTMENT OF ELECTRONICS AND ELECTRICAL  
COMMUNICATION ENGINEERING**

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# **THEORY**

## **Introduction**

In communication systems, modulation refers to the technique of modifying a carrier signal so that it can effectively transmit information over long distances. This process enhances signal strength and reliability during transmission.

Different modulation techniques are used depending on the requirement, such as Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM).

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## **Angle Modulation**

Angle modulation is a method in which the angle parameter (frequency or phase) of the carrier wave is varied according to the message signal. It is mainly classified into:

- Frequency Modulation (FM)
  - Phase Modulation (PM)
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## **Frequency Modulation (FM)**

In Frequency Modulation, the frequency of the carrier signal changes in accordance with the instantaneous amplitude of the message signal.

Unlike amplitude modulation:

- The amplitude remains constant
- The frequency varies
- The phase also changes accordingly

When the message signal is zero, the carrier frequency remains at its original value. As the message signal increases, the carrier frequency increases, and when the message signal decreases, the carrier frequency decreases.

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## **Frequency Deviation**

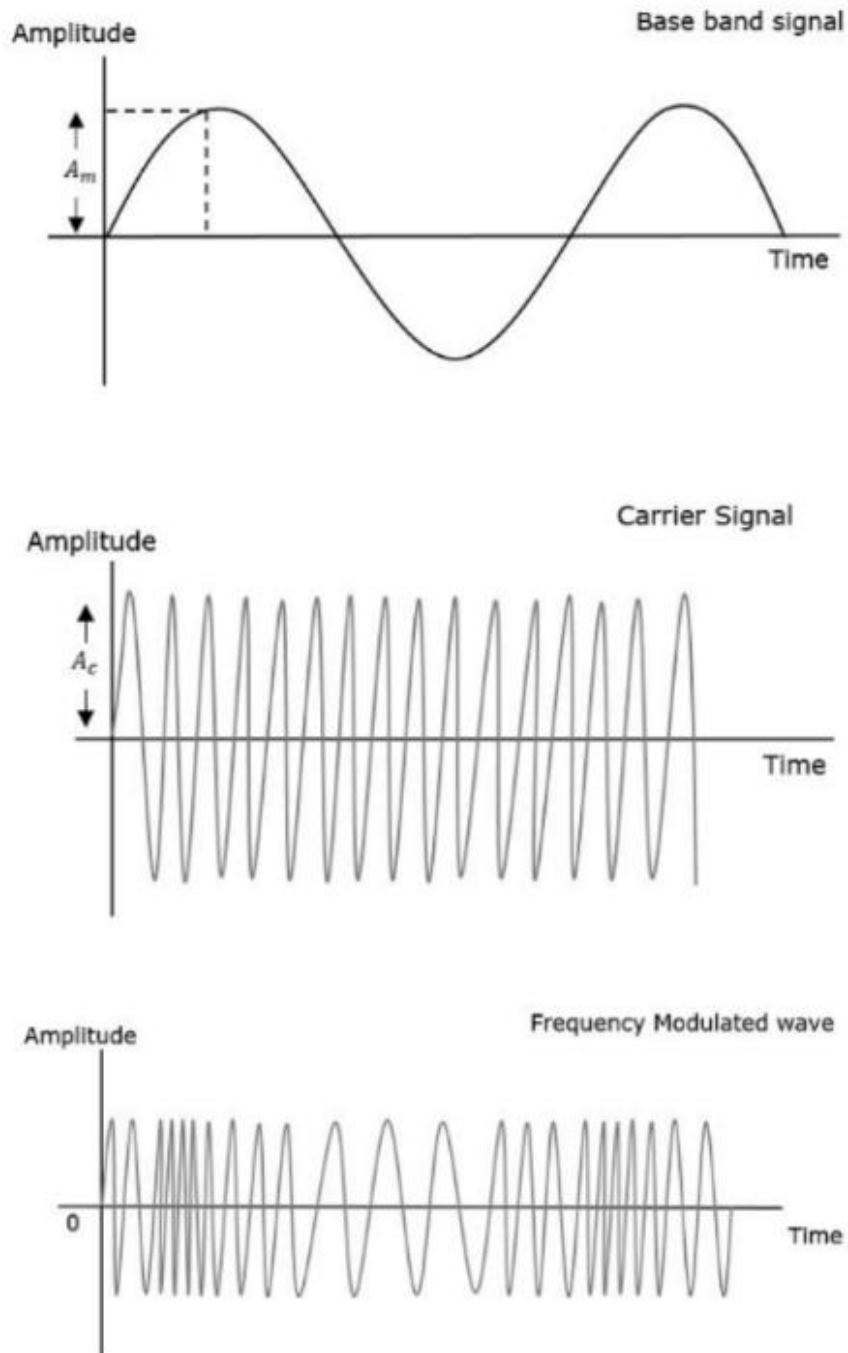
Frequency deviation is defined as the maximum shift of the carrier frequency from its central (unmodulated) value due to the message signal.

It is an important parameter in FM systems and is given by:

$$\Delta f = |f_{max} - f_c|$$

where:

- $f_{max}$  = maximum frequency
  - $f_c$  = carrier frequency
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**Fig : Frequency Modulation Waveform**

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### **Frequency Demodulation**

Demodulation is the process of retrieving the original message signal from the modulated FM wave.

In FM systems, demodulation is used to extract audio or information signals accurately. It ensures that the received signal closely matches the transmitted signal with minimal distortion.

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## Mathematical Expressions

### FM Signal Equation

$$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t) dt]$$

Where:

- $A_c$  = amplitude of carrier
  - $f_c$  = carrier frequency
  - $k_f$  = frequency sensitivity
  - $m(t)$  = message signal
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## Instantaneous Frequency

$$f_i(t) = f_c + k_f m(t)$$

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## Modulation Index

$$\beta = \frac{\Delta f}{f_m}$$

Where:

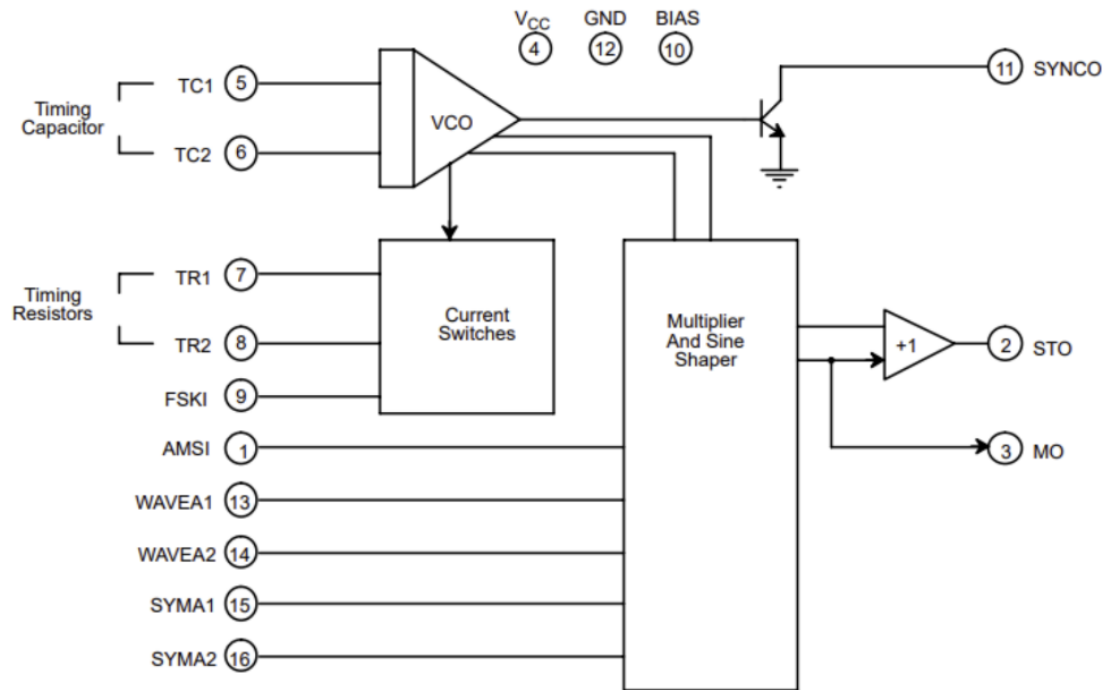
- $\Delta f$  = frequency deviation
  - $f_m$  = message frequency
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## IC XR-2206 Description

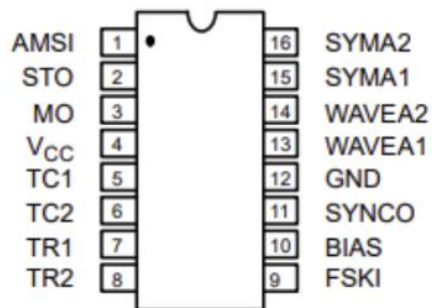
The XR-2206 is a versatile function generator IC used to generate FM signals. It consists of the following internal sections:

- Voltage Controlled Oscillator (VCO)
- Analog multiplier and sine shaping circuit
- Buffer amplifier
- Current switching network

The output frequency depends on the current flowing through timing resistors connected to specific pins.



**Fig : Block Diagram of XR2206**

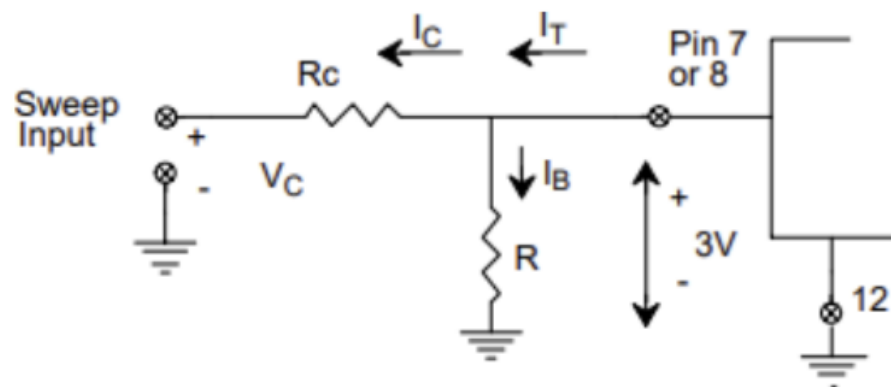


**Fig : Pin Diagram of XR2206**

## **Frequency Control and Sweep**

The oscillation frequency is directly proportional to the timing current  $I_T$ .

- Timing pins are internally biased at a fixed voltage.
- By applying a control voltage, the output frequency can be varied.
- This enables frequency sweep and modulation.



**Fig : Circuit for frequency sweep**

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### **EXPERIMENT OBJECTIVE**

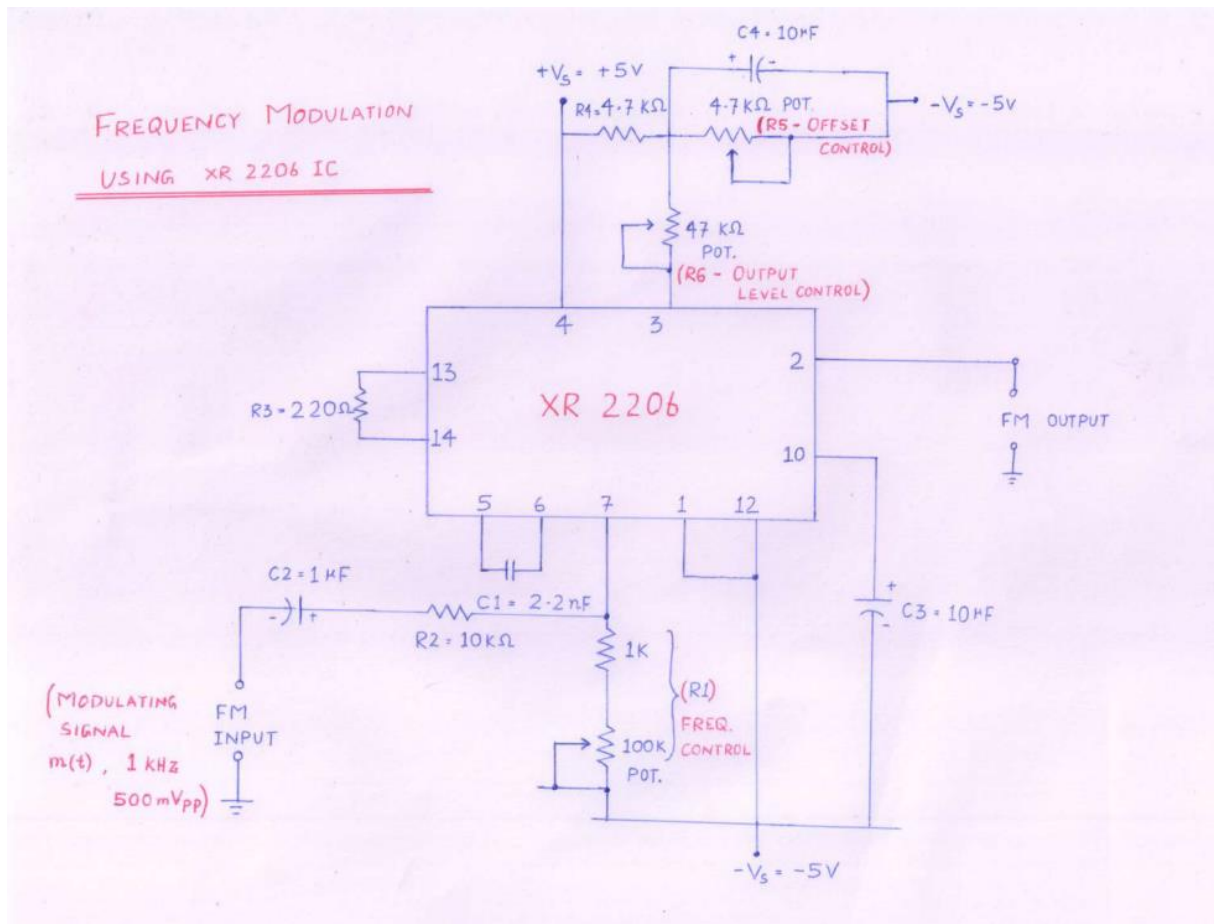
To generate frequency modulated signals using XR-2206 IC and observe the output in both time and frequency domains using a Digital Storage Oscilloscope (DSO).

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### **APPARATUS REQUIRED**

- XR-2206 IC
  - Resistors and capacitors
  - Breadboard
  - Connecting wires
  - Digital multimeter
  - Signal generator
  - Digital Storage Oscilloscope
  - DC power supply (+12V, 0V, -12V)
  - Potentiometer
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## PROCEDURE

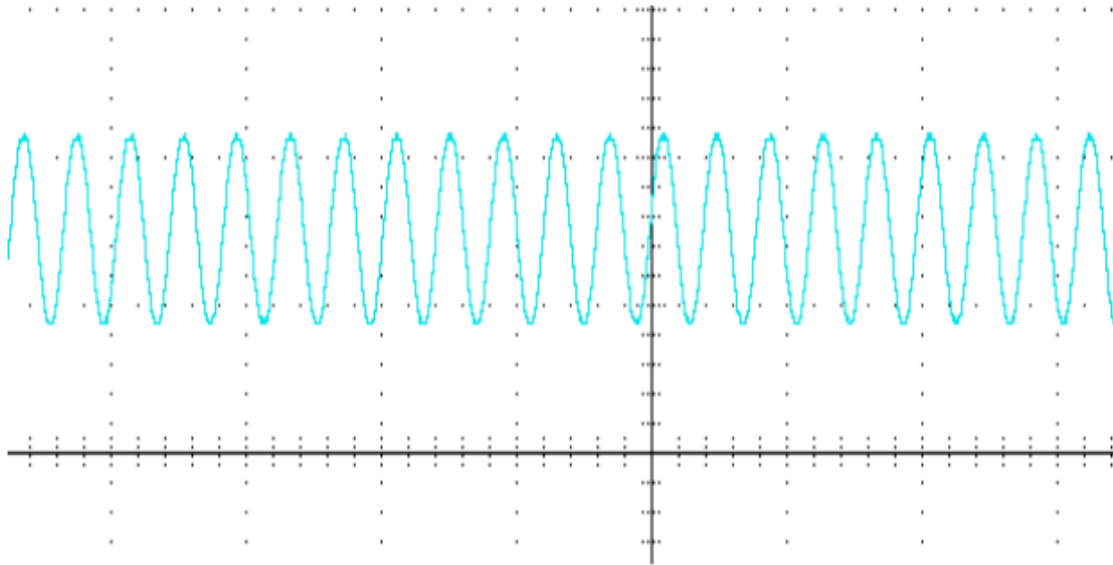


**Fig: Frequency Modulation using XR-2206**

1. Assemble the circuit according to the given diagram.
2. Keep the modulating signal zero (grounded) and adjust the potentiometer to obtain a 10 kHz carrier signal.
3. Apply a sinusoidal modulating signal (e.g., 1 kHz).
4. Observe the FM waveform on the oscilloscope.
5. Measure maximum and minimum frequencies.
6. Calculate frequency deviation and modulation index.
7. Observe the frequency spectrum (FFT).
8. Repeat the process for different input amplitudes.

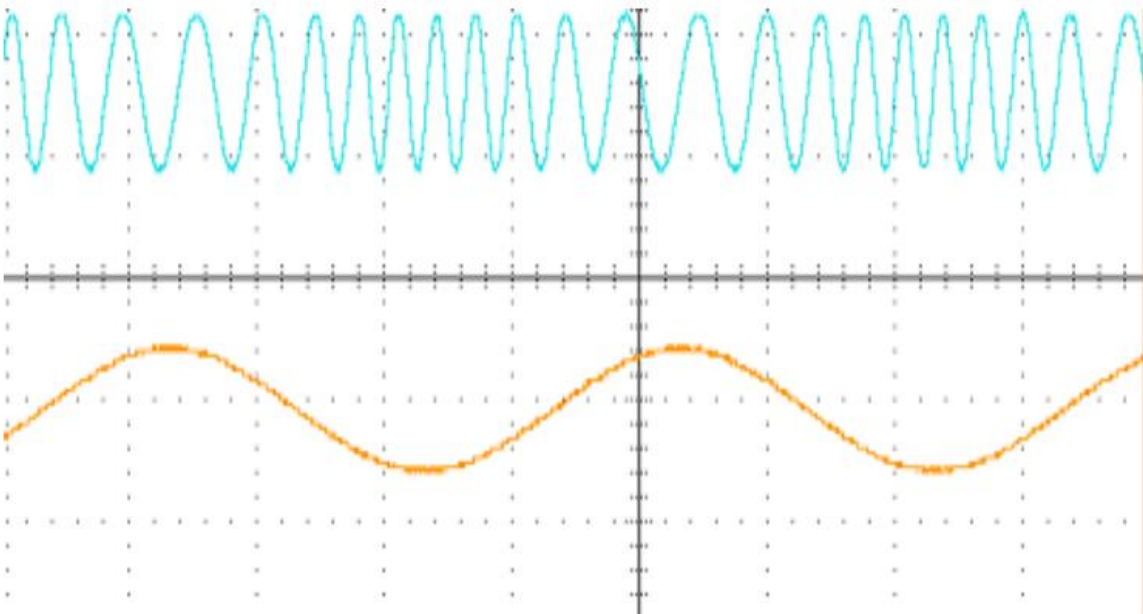
## OBSERVATIONS

- **Carrier signal observed at 10 kHz**



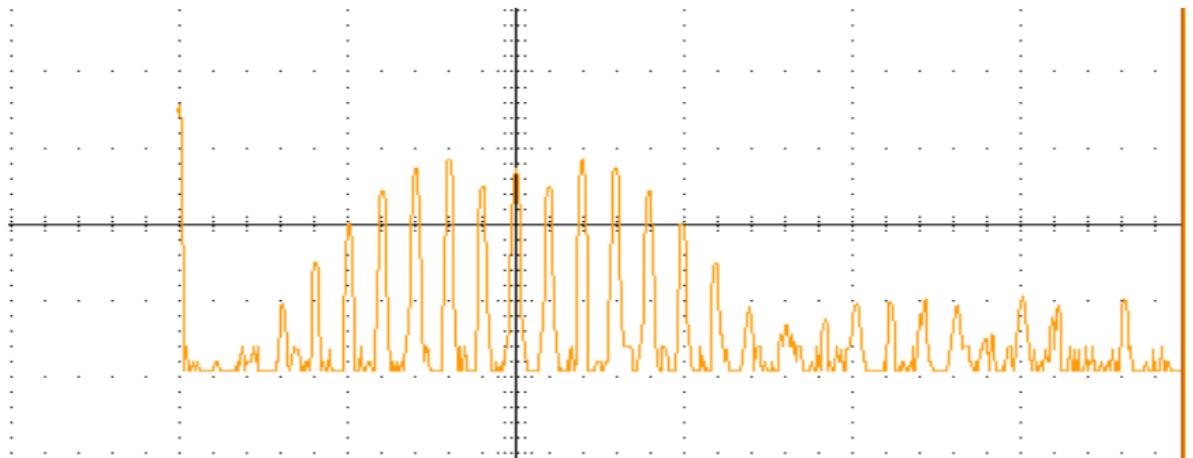
**Fig: Expected Carrier Signal  $f_c = 10\text{KHz}$ , 4Vpp**

- FM waveform varies with input amplitude



**Fig: Expected Frequency Modulation Output Waveform and modulating signal**

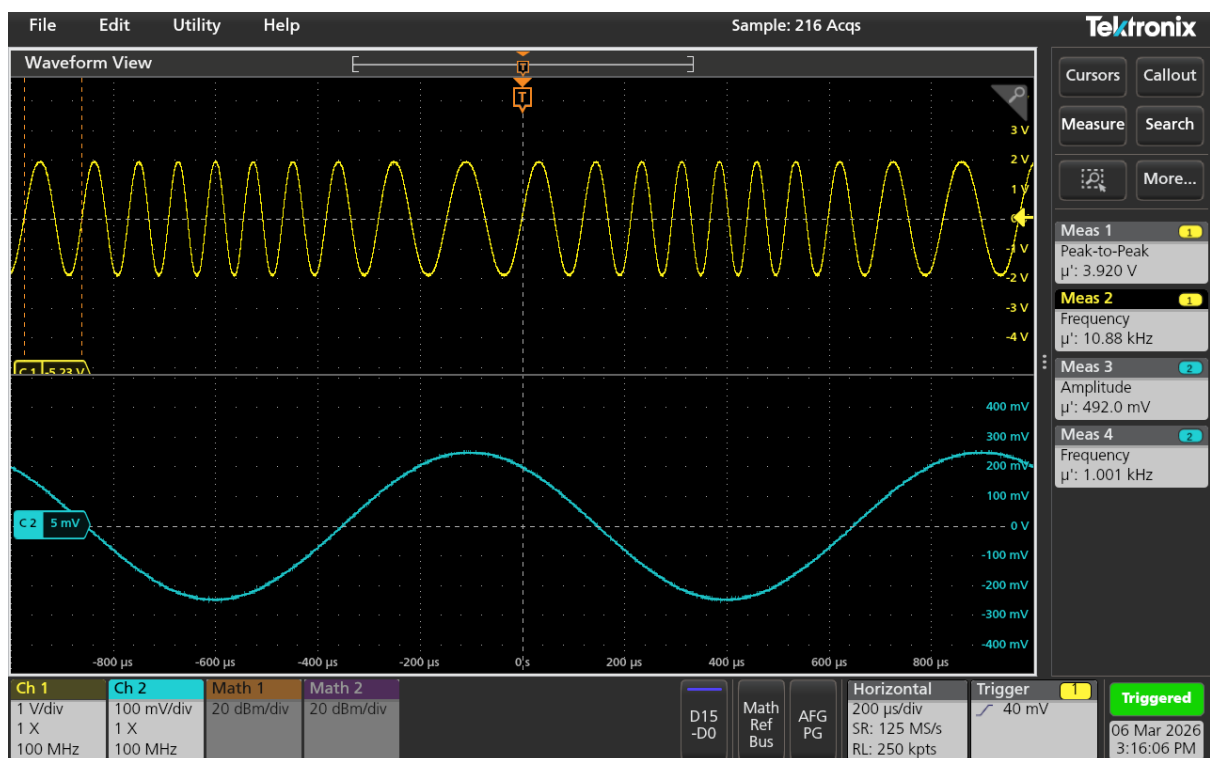
- Frequency spectrum shows carrier and sidebands



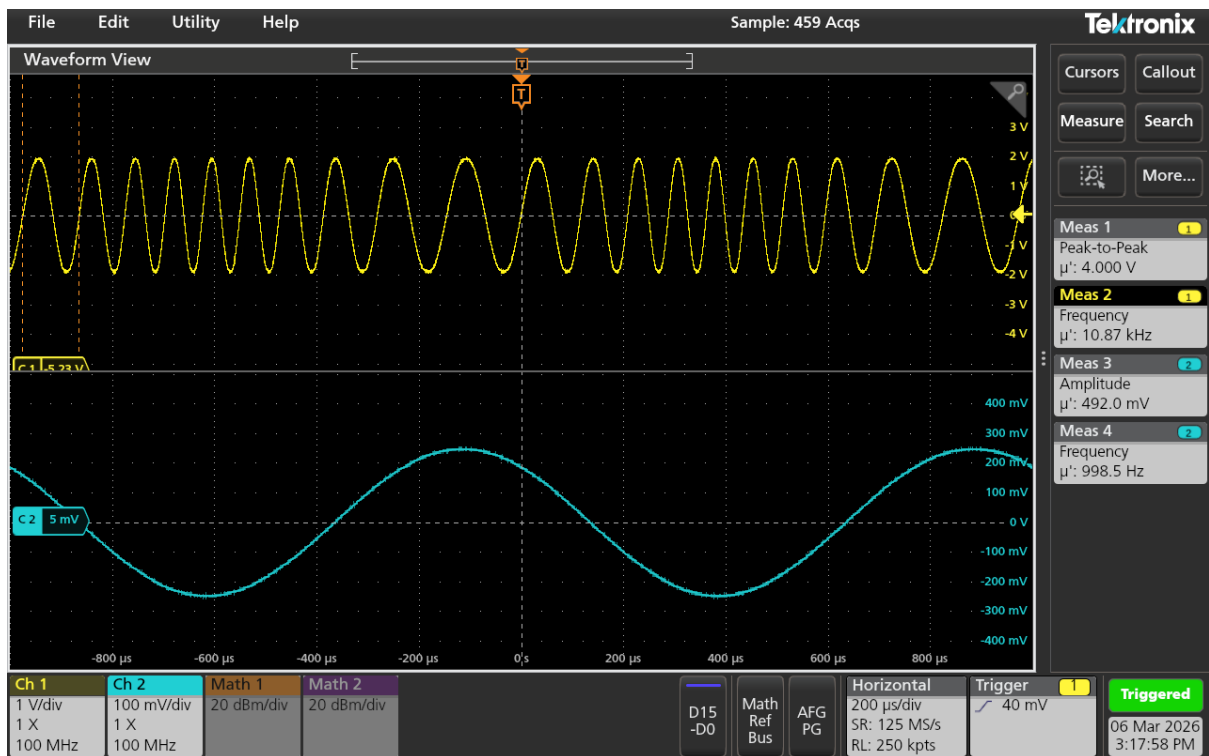
**Fig: Expected Frequency Spectrum of FM**

- Increasing message amplitude increases frequency deviation

## **CALCULATIONS AND RESULTS**



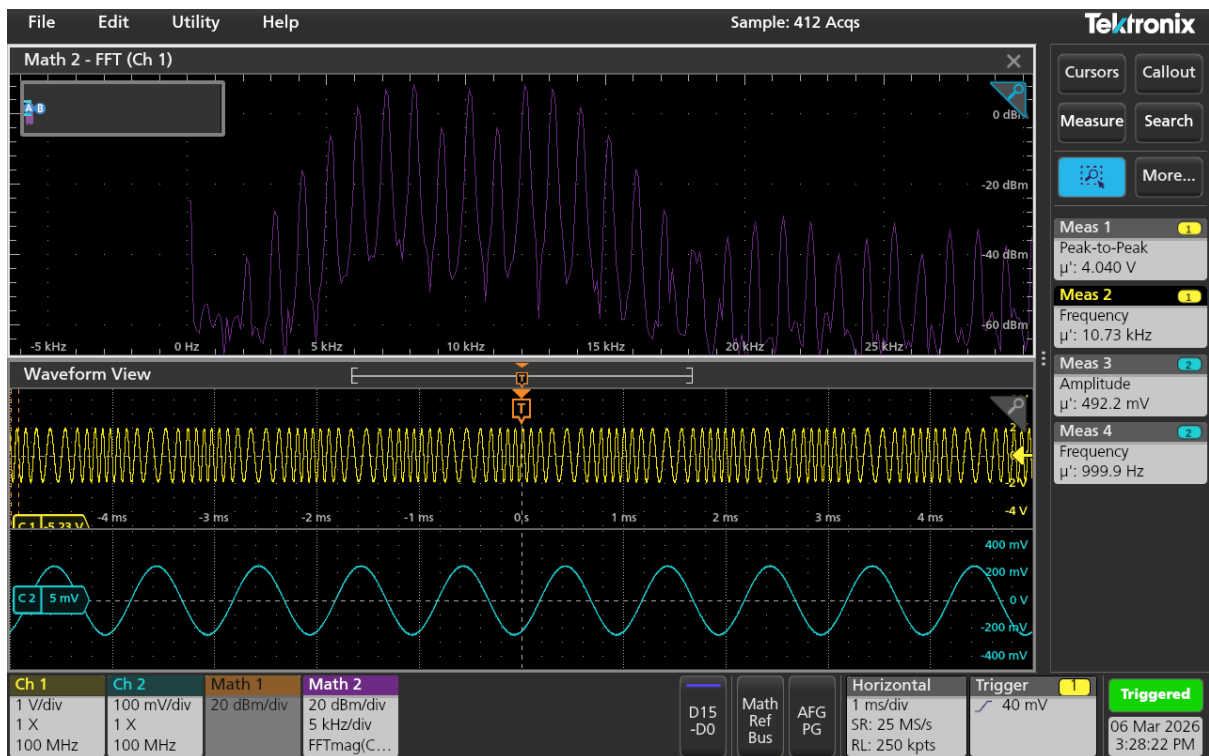
**Fig: Sinusoidal Carrier Signal of Frequency of 4Vpp and 10kHz**



**Fig: FM modulated signal at 500mVpp 1KHz**



**Fig: FFT Spectrum of FM modulated signal at 500mVpp 1KHz**



**Fig: FFT Spectrum of FM modulated signal at 500mVpp 1KHz with clear spikes**

## Maximum and Minimum Frequency

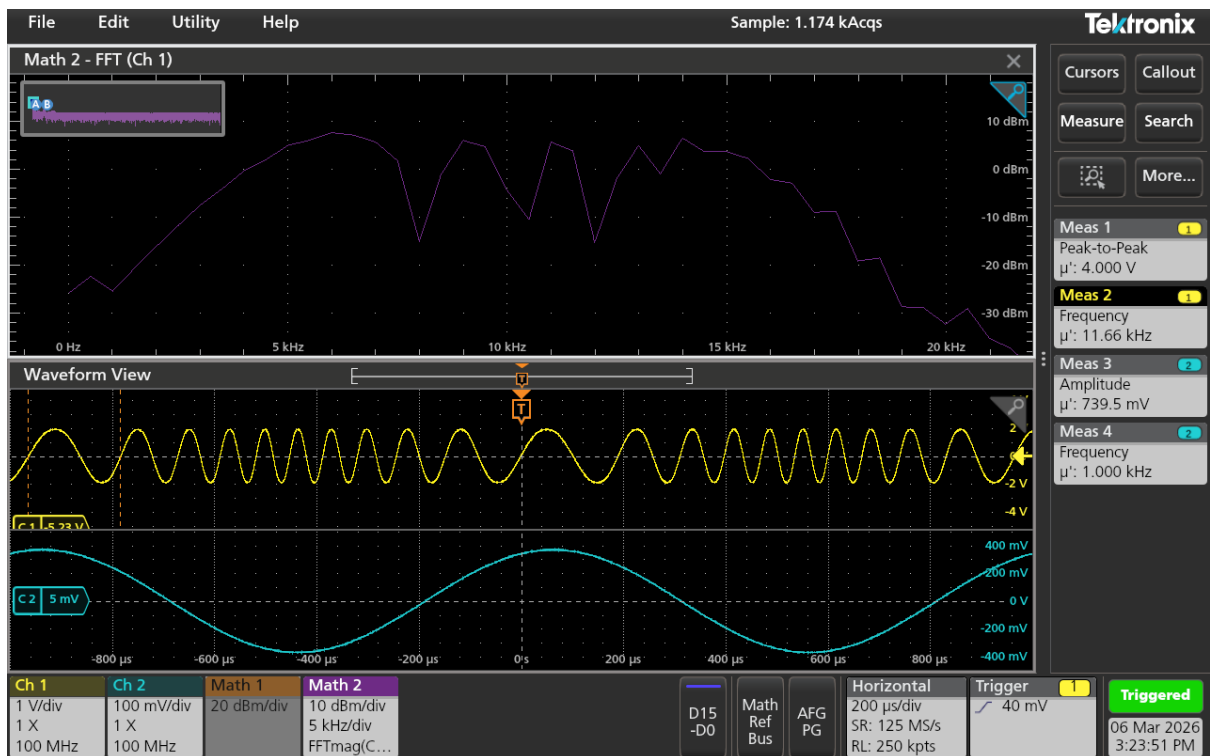
Measured from DSO in both time and frequency domain.



**Fig: Calculating and cross checking the Minimum Frequency ( $F_c - F_m$ )**



**Fig: Calculating and cross checking the Maximum Frequency ( $F_c + F_m$ )**



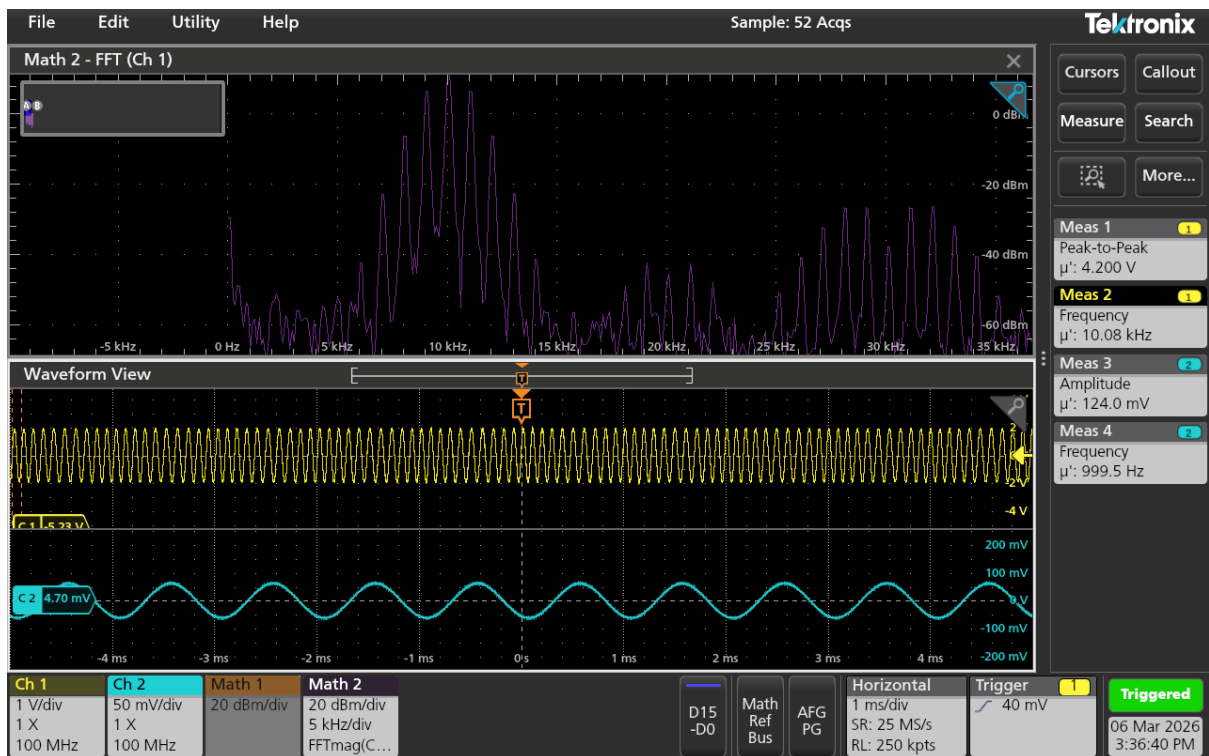
**Fig: FFT Spectrum of FM modulated signal at 750mVpp 1KHz**



**Fig: FFT Spectrum of FM modulated signal at 750mVpp 1KHz with clear spikes**



**Fig: FFT Spectrum of FM modulated signal at 250mVpp 1KHz with clear spikes**



**Fig: FFT Spectrum of FM modulated signal at 125mVpp 1KHz with clear spikes**



**Fig: FFT Spectrum of FM modulated signal at 60mVpp 1KHz with clear spikes**

| <u>Voltage (Amplitude of the message signal in mVpp)</u> | <u>F<sub>max</sub> (in time domain) (in KHz)</u> | <u>F<sub>min</sub> (in time domain) (in KHz)</u> | <u>F<sub>max</sub> in frequency domain (in KHz)</u> |
|----------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| 750                                                      | 11.05                                            | 8.423                                            | 11.53                                               |
| 500                                                      | 10.89                                            | 7.936                                            | 12.64                                               |
| 250                                                      | 11.28                                            | 7.589                                            | 12.89                                               |

### Frequency Deviation

$$\Delta f = |f_{max} - f_c|$$

F<sub>c</sub> = 10KHz, so at 500mVpp the deviation is 0.89

### Modulation Index

$$\beta = \frac{\Delta f}{f_m}$$

At 500mVpp the Modulation Index is 0.89

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## DISCUSSION:

### 1. Nature of Angle Modulation

Angle modulation (FM) is a **nonlinear process**, making both its analysis and implementation more complex than Amplitude Modulation (AM). Unlike AM, FM has a **theoretically infinite bandwidth** due to the presence of infinite sidebands, although in practice only the significant sidebands are considered.

### 2. Noise and Power Characteristics

Since the **carrier amplitude remains constant**, FM is less affected by noise, as noise primarily alters amplitude rather than frequency. This results in **better noise immunity** compared to AM. Additionally, FM maintains **constant transmitted power**, independent of the modulating signal amplitude.

### 3. Properties of FM Signal

- The observed FM waveform verifies key theoretical properties:
- **Constant amplitude**
- **Nonlinear modulation process**
- **Irregular zero crossings due to frequency variation**
- **Difficulty in directly visualizing the message signal**
- **Increased bandwidth requirement for improved performance**

### 4. XR2206 IC Behavior

The XR2206 function generator exhibits an **inverted FM response**:

- Increase in control voltage → **decrease in frequency**

This occurs due to the internal **Voltage-Controlled Oscillator (VCO)** design, where frequency is inversely proportional to the applied control voltage.

### 5. FFT and Bessel Function Analysis

- The FFT spectrum shows **equally spaced sidebands at 1 kHz intervals**, corresponding to the modulating frequency.
- Sidebands occur at:  $f_c \pm n f_m$
- Amplitudes follow **Bessel functions of the first kind**
- Spectrum is **symmetric about the carrier frequency**
- Higher-order sidebands decay as  $n > \beta n$
- This confirms the theoretical representation of FM using **Bessel function expansion**.

### 6. Frequency Deviation and Distortion

For the experiment:

- Carrier frequency  $f_c = 10\text{kHz}$
- Modulating signal  $f_m = 1\text{kHz}$ ,  $500\text{mVpp}$

The observed **modulation index** is:

$$\beta = 0.89$$

Thus, the **peak frequency deviation** is:

$$\Delta f = \beta \cdot f_m = 0.89 \times 1\text{kHz} = 0.89\text{kHz}$$

This indicates **moderate FM**, where only a few significant sidebands are present.

In earlier observations, distortion occurred when the **frequency deviation exceeded the carrier frequency**, leading to **loss of information and spectral imbalance**. This highlights the importance of maintaining proper **frequency headroom**.

## 7. FM and PM Relationship

Applying a **differentiator** to the message signal produces **phase modulation (PM)**. This shows that FM and PM are closely related, with FM depending on the integral and PM on the instantaneous value of the message signal.

## 8. Circuit Observation

When the potentiometer was used with an **open terminal**, the circuit showed unstable behavior. The floating terminal acted as an **antenna**, introducing noise and causing improper carrier generation. Proper connection removed this issue and stabilized the output.

## 9. Advantages and Limitations of FM

### Advantages:

- High noise immunity
- Better signal quality
- Constant transmitted power

### Disadvantages:

- Large bandwidth requirement
- Increased circuit complexity
- Higher implementation cost

## 10. AM vs FM

| Feature        | AM     | FM      |
|----------------|--------|---------|
| Noise Immunity | Low    | High    |
| Bandwidth      | Low    | High    |
| Signal Quality | Poor   | Good    |
| Complexity     | Simple | Complex |